A FINITE MODULE ADMITS A FILTRATION WHOSE SUCCESSIVE QUOTIENTS ARE CYCLIC

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance, as required for this source: the first line of the pinned chapter file is \input{preamble}, and the native preamble was not staged with the source, so no preamble text lives inside the chapter. The retained preamble used here is the project's single Stacks-wide preamble (the same 217-line / 8116-byte file that the already published chow and etale-cohomology cards use), copied mechanically line for line from sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Not one character was changed, no package was added, no macro was defined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The same-chapter pointers are located at their \label lines inside the pinned chapter file; the cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. This card is the test case for section 13.2: its counted statement contains a slogan environment, which the shared preamble defines as a comment environment in lines 8-13, so the preamble spans must include those lines; the generator now emits lines 8-18 plus the runtime tail of the file instead of the previously hardcoded 14-18.

Editorial note. The counted proof contains no \ref and no \cite, so the card carries no navigation stub and no bibliography.

Allowed standard prerequisites (editorial).

- The number of generators of a quotient by a cyclic submodule decreases by one. Used inside the counted proof when the induction hypothesis is applied to M/M'. Supplies the induction step: M/M' has r - 1 generators, so the hypothesis gives a filtration there.
- A cyclic module is isomorphic to R modulo its annihilator ideal. Used inside the counted proof at its last sentence. Identifies the cyclic submodule M' = Rx_1 with R/I_1 for I_1 the annihilator of x_1.

Lemma 0.1. *Let R be a ring, and let M be a finite R-module. There exists a filtration by finite R-submodules*

$$0 = M_0 \subset M_1 \subset \ldots \subset M_n = M$$

such that each quotient M_i/M_{i-1} is isomorphic to R/I_i for some ideal I_i of R.

Proof. By induction on the number of generators of M. Let $x_1, \ldots, x_r \in M$ be generators. Let $M' = Rx_1 \subset M$. Then M/M' has $r-1$ generators and the induction hypothesis applies. And clearly $M' \cong R/I_1$ with $I_1 = \{f \in R \mid fx_1 = 0\}$. □

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.